STUDY · NEGATIVE RESULT · 16 OCT 2026

A pre-registered test of volatility-scaled momentum failed its own kill criteria at five positions

This page reports a negative research result. It is not a strategy, a recommendation or a signal, and nothing on it says what anyone should hold.

The mechanism passed on 45.7 years of survivorship-free factor data. The version a small account could hold, five long positions with costs charged, failed: no specification passed all four of its kill criteria, so no money was committed.

Pre-registered 10 Oct 2026 · Run 10 Oct 2026 · Re-run and published 16 Oct 2026 · Script: volmom/ · Data: Kenneth French data library; S&P 500 membership history (fja05680/sp500, MIT); Yahoo monthly closes

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Published whether or not the result was interesting. This one was not the result its author hoped for: the test was set up to decide whether a strategy would be used, and it said no.

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The short version

Volatility-scaled momentum is a published idea (Barroso & Santa-Clara, 2015): take the classic momentum factor and shrink exposure when its recent volatility is high. Before any return was computed, I wrote down three specifications, the sample splits, a cost model and seven numeric kill criteria.

Part A, the mechanism, passed. On 45.7 years of survivorship-free factor data the specification had never been fitted on, the volatility-scaled version had a Sharpe ratio of **0.78** (t = **5.25**) against **0.37** for plain momentum, and it removed most of the crash risk in the return distribution.

Part B, the version a small account could hold, failed. With five long positions, a minimum price and costs charged, **no specification passed all four of its kill criteria.** Over the same 2013–2026 holdout, simply holding the S&P 500 index fund (SPY) had a higher Sharpe ratio (**1.06**) than any of the three specifications (best **0.71**). The two specifications that earned high returns would have hit a 10% drawdown limit **11 and 9 times** in 13.75 years. Under the pre-registration, that meant no money was committed and the result was to be published.

What was tested, and why

The question had two parts, and the second was the one that mattered:

1. Does scaling momentum by its own recent volatility improve risk-adjusted returns at all?

2. Does it survive a small retail account's constraints: five positions, long only, real costs, and a 10% drawdown limit?

The academic literature answers the first, on long-short portfolios of hundreds of stocks. It does not answer the second. A pre-registered test with kill criteria set in advance was the way to get an answer that could not be adjusted after seeing it.

What was fixed before any return was computed

	Specification	Part A (factor)	Part B (five positions)
S1	Plain momentum (control)	Momentum factor, monthly	Top 5 by 12-1 month return, equal weight
S2	Volatility-scaled	Scale by 0.12 ÷ realised volatility	Same scaling on total exposure
S3	Trend filter	Hold only when the market index is above its 10-month average	Hold a name only above its own average

- **Ranking:** 12-1 momentum, the return from 12 months ago to 1 month ago, skipping the latest month.
- **Parameters:** 12% target volatility, 126-trading-day window. These are Barroso & Santa-Clara's own values and were not tuned. No parameter sweep was allowed.
- **Significance bar:** three specifications, Bonferroni-corrected, so nothing counts below **|t| = 2.394**.
- **Splits, holdout evaluated once:** Part A in-sample 1927–1980, holdout **Jan 1981 – Aug 2026 (548 months)**. Part B in-sample 1997–2012, holdout **Jan 2013 – Sep 2026 (165 months)**.

Data and period

- **Part A:** Kenneth French's momentum factor and market data, built from CRSP, which keeps delisted firms, so this part has no survivorship bias. 1,196 monthly observations from January 1927 and 26,216 daily observations from November 1926, both to August 2026; no rows dropped in parsing.
- **Part B:** point-in-time S&P 500 membership from `fja05680/sp500` (MIT), 2,720 dated snapshots; monthly closing prices from Yahoo. Constraints: 5 positions, long only, minimum price $5, and a 0.20% cost charged on each side of every trade.

Method

Part A uses the published factor directly. Part B rebuilds the strategy from individual stocks each month, with costs charged on the fraction of the portfolio replaced. Three conventions are worth knowing before reading the tables:

- **The Sharpe ratio is not annual return ÷ annual volatility here.** Annual return compounds the monthly mean; the Sharpe ratio scales it linearly. The two drift apart as the mean grows, which is why S2's row in Part A (13.31% return, Sharpe 0.78) does not divide neatly.
- **Realised volatility is uncentred**, `sqrt(252 × mean(r²))`, which is Barroso & Santa-Clara's published formula. The centred version moves S2's holdout Sharpe from 0.777 to 0.776.
- **Kurtosis is raw, not excess** (a normal distribution has 3).
- **The 10% drawdown limit was measured, not applied.** The pre-registration asked whether the holdout *would* trip it more than three times, so trips are counted on the uninterrupted return series and trading never actually stops. That is why a −42.9% drawdown can appear in Part B.

Results

Part A: the mechanism, survivorship-free (holdout 1981–2026, n = 548 months)

Specification	Annual return	Sharpe	Max drawdown	Worst month	t
S1 plain momentum	5.75%	0.37	−57.8%	−34.4%	2.49
S2 volatility-scaled	**13.31%**	**0.78**	−27.8%	−20.6%	**5.25**
S3 trend filter	4.99%	0.47	−26.0%	−12.6%	3.18

The crash risk is visible in the shape of the returns. Plain momentum's holdout skew is **−1.43** with kurtosis **13.22**. Volatility-scaled: skew **+0.05**, kurtosis **5.07**.

Implementation check against Barroso & Santa-Clara's published figures for 1927–2011. This code's figures are for November 1927 – December 2011, the paper's window after the 10-month warm-up the trend filter needs; they come from the independent implementation in `verify/`:

	This code	The paper
S1 Sharpe	0.48	0.53
S2 Sharpe	0.93	0.97
S2 max drawdown	−42.6%	−45.2%
S2 worst month	−29.0%	−28.4%
S1 worst month	−52.6%	−79.0%

S2 reproduces closely. S1's worst month does not. The likely cause is that the paper builds its factor from six size-by-momentum portfolios while

this code uses French's published factor. That has not been verified, and the gap stays on the record.

Part A kill criteria

	Criterion	Measured	Verdict
K1a	S2 holdout Sharpe ≥ 0.50	0.78	Pass
K1b	S2 beats S1 by ≥ 0.15 Sharpe	0.41	Pass
K2	Probability of backtest overfitting ≤ 0.50	0.00 to 0.76	**Indeterminate**
K3	Deflated Sharpe "≤ 0"	0.876 / 1.000 / 0.973	**Cannot fire**

K2 and K3 are defects in the pre-registration, not results. K2 never said which return matrix to use: in-sample gives **0.7638** (fail), holdout **0.0000** (pass), the common sample **0.0828** (pass). With only three specifications the statistic has almost no resolution, so it is reported as a range. K3 was written as "≤ 0", but the deflated Sharpe is a probability between 0 and 1, so it could never have fired. A conventional > 0.95 threshold would pass S2, but that threshold was not pre-registered and is not used here as a pass.

Part B: five positions, long only, costs charged (holdout 2013–2026, n = 165 months)

Specification, 0.20% per side	Annual return	Sharpe	Max drawdown	Times a 10% limit would trip
S1 plain momentum	25.49%	0.71	−37.3%	**11**
S2 volatility-scaled	10.57%	0.66	−24.6%	**2**
S3 trend filter	25.69%	0.71	−42.9%	**9**
SPY, held throughout (benchmark)	16.05%	1.06	−23.9%	3

Every Part B return is an upper bound, for the survivorship reason below.

In-sample (1997–2012) was no better: **no specification reached t = 2**, and SPY's Sharpe of 0.42 matched S2's 0.43.

Part B kill criteria: no specification passes all four

	K4 Sharpe ≥ 0.30	K5 beats SPY	K6 ≤ 3 limit trips	K7 breakeven cost ≥ 0.50% round trip
S1	0.71 Pass	Pass	**11 Fail**	0.868% Pass
S2	0.66 Pass	**Fail**	2 Pass	**0.467% Fail**
S3	0.71 Pass	Pass	**9 Fail**	0.866% Pass

S1 and S3 earn the returns and fail on drawdown. S2 stays within the limit and loses to the benchmark. No specification passes everything, so under the pre-registration nothing was traded.

What this does not show

1. **It is not a strategy, a recommendation or a signal, and it is not a recommendation to hold SPY or anything else.** It is a negative research result. SPY is the benchmark the pre-registration named. Beating a strategy over one 13.75-year window is a fact about that window.
2. **Free price data cannot see the stocks that failed.** Of 503 current index members, 501 had prices (**99.6%**); of 706 departed members, only 261 did (**37.0%**). At rebalance dates, median index coverage was **52.6% in-sample and 84.8% in the holdout**, as printed by the public script on fresh downloads on 16 October 2026 (an earlier README figure of 85.0% could not be reproduced; see Versions). The missing names are disproportionately the ones that went bust, so Part B returns are biased upward. Pricing delistings at 0%, −30% (Shumway 1997) or −100% gives identical results, because the missing names never enter the panel at all.
3. **The "zero throttling" check on the price download is weak.** The fetch logged no rate-limiting, but the detector cannot see a batch that silently comes back empty. Some of the missing departed names could be missing for that reason rather than because the source lacks them.
4. **2013–2026 was an unusual period for large-cap momentum.** A top-5 screen on US large caps over that window is largely a concentrated bet on a few mega-cap technology names. 25% a year with a −37% drawdown and 11 limit trips looks like a sector bet, not a durable edge.
5. **Part A is not tradeable as measured.** It is a long-short factor across hundreds of stocks with no costs. Every difference between that and five long positions cuts the same way.
6. **Two forced deviations in Part B.** The pre-registration specified daily windows but Part B has monthly prices, so S2 uses 12 monthly returns for volatility (noisier, leverage capped at 3×) and S3 uses a 10-month average instead of 200 days. Neither was chosen after seeing a result.

7. **Only Part A has a second implementation.** Part A was rebuilt from the pre-registration text by code written without reading the original computation, and every Part A figure matched. Part B has no independent reimplementation.

Reproduce it

Folder: `volmom/` in `donttrade/disposition-effect`.

```
pip install -r requirements.txt
python reproduce_volmom.py
```

Part A takes about a minute. Part B fetches monthly closes for 1,209 tickers on first run (about ten minutes), then caches them. Both download their own data; no data is shipped in the repository.

On 16 October 2026 the public script was re-run from a clean directory, downloading all of its own data. Every Part A and Part B figure on this page matched its output.

Since 16 October 2026 the script prints each breakeven cost both per side (0.434%, 0.234%, 0.433%) and as a round trip (0.868%, 0.467%, 0.866%), and the K7 verdicts S1 pass, S2 fail, S3 pass, matching this page. Its Part B banner says the 10% limit is counted, not applied (see Versions).

The independent Part A reimplementation, which also produces the K2 range, is in `volmom/verify/`:

```
pip install pandas numpy scipy
python fetch.py
python volmom.py
```

References

- Barroso, P. & Santa-Clara, P. (2015). "Momentum has its moments." *Journal of Financial Economics* 116(1), 111–120.
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- Bailey, D. & López de Prado, M. (2014). "The Deflated Sharpe Ratio." *Journal of Portfolio Management* 40(5).
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- Kenneth French data library. fja05680/sp500 (MIT).

Methods and negative results, yes. Live signals, never.
Again: this is a negative research result, not a strategy, a recommendation or a signal. Nothing here says what anyone should buy, sell or hold.

Versions

Date	What changed
10 Oct 2026	Pre-registration written before any return was computed. Part A and Part B run.
10 Oct 2026	Correction: the first write-up said a look-ahead filter in the Part B portfolio builder inflated returns. Measured afterwards, it changed nothing: there were 0 delistings in 1,775 position-months. The claim was withdrawn.
13 Oct 2026	Write-up and code prepared for `volmom/`. Added the three conventions in Method, the K2 range and the K3 defect, and the independent Part A reimplementation.
After 13 Oct 2026	Three corrections to `volmom/README.md`: the 10% drawdown limit is measured, not applied; costs are 0.20% per side (0.40% round trip), and the pre-registration's own wording allowed both readings, a third defect in it; the heading "every specification fails at least two" was false, because per-side breakevens had been compared against a round-trip floor. Doubled to round trips, S1 and S3 pass K7. The conclusion did not change: no specification passes all four.
16 Oct 2026	Published on this site. Re-run from a clean directory with fresh downloads: every figure matched. The script's labels now match this page: it prints breakevens per side and round trip, the K7 verdicts S1 pass, S2 fail, S3 pass, and a banner saying the 10% limit is counted, not applied; its downloads send an explicit User-Agent. No computed number changed. S2's round-trip breakeven is 0.467%, as the script prints. Holdout median coverage is 84.8%, as printed on two fresh fetches; an earlier README figure of 85.0% could not be reproduced. See Corrections.

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